Resolution Risk in Prediction Markets

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Abstract

Prediction-market prices are commonly interpreted as probabilities of underlying events. That interpretation assumes that the exchange will map the realized event into the intended contract payoff. This project studies the possibility that prices also reflect *resolution risk*: uncertainty that a contract will be resolved in a manner that differs from traders' economic interpretation of its terms. We construct high-precision matches between Polymarket and Kalshi contracts and study two 2025 controversies involving Polymarket's UMA-based resolution process. The primary outcomes measure each price's absolute distance from 0.5 and Polymarket's distance relative to Kalshi's. Preliminary estimates do not support the simple hypothesis that a public resolution controversy causes prices broadly to attenuate toward 0.5. The exercise is informative, but it is not yet a causal estimate: the available data, matching procedure, event windows, weights, and standard errors all require further work.

1 Introduction

A binary prediction-market price is often read as the market's probability that an event will occur. Even in an ideal contract this interpretation requires assumptions about beliefs, risk preferences, wealth, and trading frictions (Manski, 2006; Wolfers and Zitzewitz, 2006). In practice, it also requires confidence in the institution that converts the realized event into a payoff. A trader may believe that an event occurred while assigning positive probability to the exchange resolving the contract the other way. The contract price then reflects both beliefs about the event and beliefs about resolution.

This project studies that second component, which we call *resolution risk*. The institutional comparison is between Polymarket and Kalshi. During the period studied here, the matched Polymarket contracts were ultimately backed by UMA's optimistic-oracle and dispute process, either directly or through Polymarket's negative-risk adapter. Kalshi resolves contracts internally: the exchange applies its own contract rules and settlement procedures rather than relying on UMA or another tokenholder-based dispute mechanism. Equivalent contracts on the two exchanges are therefore exposed to similar event news but different resolution institutions.

We focus on two public controversies: the March 24, 2025 Ukraine mineral-rights resolution and the June 30–July 8, 2025 Zelensky-suit episode. The research question is whether these episodes changed the informativeness of Polymarket prices relative to matched Kalshi prices. Our main hypothesis is attenuation. If a controversy adds uncertainty about which side of a contract will be paid, then a Polymarket price should move toward 0.5 relative to an otherwise equivalent Kalshi price. This prediction is deliberately symmetric: the relevant object is not whether the average Yes price rises or falls, but whether Polymarket prices become less extreme.

2 Related Literature and Contribution

The project is closest to three literatures. First, prediction-market prices need not equal literal probabilities, even when markets aggregate dispersed information (Wolfers and Zitzewitz, 2004; Manski, 2006; Wolfers and Zitzewitz, 2006). Resolution risk supplies a distinct institutional wedge: a claim can be priced differently even when traders agree about the underlying event because they disagree about how the contract will be interpreted or enforced.

Second, work on manipulation in prediction markets asks whether strategic trading can distort prices or whether traders can change the event being forecast (Hanson, Oprea, and Porter, 2006; Ottaviani and Sørensen, 2007; Rhode and Strumpf, 2008). Our mechanism is different. The object at risk is the mapping from the event to the payoff, not necessarily the event itself or the trading price before resolution. This connects the paper to recent work on semantic ambiguity in prediction-market contracts (Rezaei Sanjabi, 2026) and to the computer-science literature on blockchain oracles, governance, and manipulation-resistant data feeds (Eskandari et al., 2021).

Third, matched contracts across exchanges resemble law-of-one-price comparisons. Limits to arbitrage, collateral constraints, segmentation, and market-specific liquidity can sustain price wedges even for closely related claims (Shleifer and Vishny, 1997; Froot and Dabora, 1999; Gârleanu and Pedersen, 2011). These forces are not merely background citations; they are competing explanations that the empirical design must address. The intended contribution is therefore narrow: to test whether public shocks to the perceived credibility of a resolution institution alter the informativeness of otherwise comparable prediction-market prices.

3 Institutions and Hypothesis

UMA uses an optimistic-oracle design. An outcome is proposed, a challenge period allows a dispute, and disputed cases can ultimately reach UMA's decentralized verification mechanism. This structure is designed to make incorrect resolution costly, but it also makes the final payoff depend on an institutional process that is distinct from the underlying event. Polymarket's negative-risk markets add an adapter layer, but the matched sample's ultimate resolver remains UMA.

Let p_{it}^P and p_{it}^K denote the Polymarket and Kalshi prices for matched contract i at time t. Define each price's extremity as

$$A_{it}^{P} = \left| p_{it}^{P} - 0.5 \right|, \tag{1}$$

$$A_{it}^{K} = \left| p_{it}^{K} - 0.5 \right|, \tag{2}$$

and define relative extremity as

$$R_{it} = A_{it}^{P} - A_{it}^{K}. \tag{3}$$

A negative change in A_{it}^{P} indicates that Polymarket moves toward 0.5 after a controversy. A negative change in R_{it} indicates attenuation on Polymarket relative to the matched Kalshi contract. The second outcome is the cleaner comparison because it removes common changes in how close the event itself is to resolution. We do not emphasize a directional Polymarket–Kalshi price difference because resolution uncertainty can impair either side of a binary contract; its sign depends on which outcome is vulnerable.

4 Data and Matching

4.1 Source data and price construction

The analysis uses the J.D. Becker archive of Polymarket and Kalshi market and trade data. For each event window, we construct observations at 5:00 a.m. and 5:00 p.m. Pacific using the most recent trade at or before the cutoff. The analysis is therefore a panel of *as-of price snapshots*, not a synchronized panel of contemporaneous quotes or order books. A price may be many hours or days old when a market trades infrequently.

This distinction is especially important for nonprice variables. The archive and pipeline do not provide aligned time series for bid–ask spreads, depth, open interest, or exchange-level participation. Polymarket volume enters as a market-level snapshot rather than a contemporaneous, event-time measure, and comparable Kalshi volume histories are not used. Consequently, the analysis cannot determine whether a price response is accompanied by a change in liquidity or participation, and its volume weights are ex post.

4.2 Deterministic matching

The matching pipeline parses each contract into a structured key containing the market family and relevant semantic fields, such as predicate, subject, object, threshold, and deadline. Aliases and economically equivalent deadlines are normalized before exact keys are joined across exchanges. The parser covers 17 contract families, including elections, office exits, leader meetings, sports, Federal Reserve decisions, policy actions, and asset-price thresholds.

The procedure begins with 5,680 Polymarket markets and 8,992 Kalshi markets. It parses 1,346 Polymarket markets and 1,979 Kalshi markets, then produces 150 deterministic matched rows representing 149 unique keys. Parsing rates are low in counts—23.7 percent for Polymarket and

22.0 percent for Kalshi—but much higher when weighted by reported volume. On-chain checks classify 15 matched Polymarket markets as direct UMA markets and 135 as resolving through the negative-risk adapter to UMA.

A more flexible matcher could improve coverage without sacrificing the accuracy of the deterministic approach by separating candidate generation from adjudication. Markets could first be segmented by broad family, date range, and named entities. Within each segment, embeddings or lexical retrieval could generate a small candidate set. A language model could then compare the full contract text and resolution rules using a fixed schema, returning a match decision, the specific rule differences, and a calibrated confidence score. Human review of accepted matches would preserve a high-precision final sample while extending the method to contracts not covered by hand-written parsers.

5 Empirical Design

For each controversy, the empirical analysis constructs a symmetric seven-day window around the event period and estimates

$$Y_{it} = \alpha_i + \beta\,\mathrm{Post}_t + \varepsilon_{it}, \tag{4}$$

where Y_{it} is either A^P_{it} or R_{it} and α_i is a matched-contract fixed effect. Observations are weighted by the available Polymarket volume measure. The mineral- rights event contributes 105 matched pairs and 3,105 observations; the Zelensky-suit event contributes 57 pairs and 2,565 observations. A narrower "resolution-risk-exposed" sample keeps elections, office exits, leader contacts, and policy actions, leaving 12 and 19 pairs for the two episodes.

Equation (4) is a pre/post fixed-effect comparison, not a modern dynamic event study or a fully specified difference-in-differences design. Kalshi enters through the construction of R_{it} rather than through an exchange-level panel with matched-pair-by- time fixed effects. The present specification also imposes a single discontinuity at the start of each episode even though the Zelensky controversy spans multiple days.

6 Preliminary Results

Table 1 reports the estimates in the conventional economics format: coefficients followed by standard errors in parentheses. Negative coefficients would support attenuation toward 0.5. The estimates are generally positive. In the full sample, the mineral-rights episode produces essentially no relative change, while the Zelensky episode is associated with a 0.47 percentage point increase in Polymarket's extremity relative to Kalshi. In the resolution-risk-exposed sample, the corresponding Zelensky estimate is a 1.47 percentage point increase. These patterns reject the simplest version of the attenuation hypothesis in this specification.

The results do not establish that resolution concerns had no effect. The hypothesized response may be concentrated in the disputed contracts themselves, in contracts near their terminal dates,

Table 1: Resolution Controversies and Price Extremity

	Mineral rights		Zelensky suit	
	$\|P-.5\|$	$\|P-.5\|-\|K-.5\|$	$\|P-.5\|$	$\|P-.5\|-\|K-.5\|$
	(1)	(2)	(3)	(4)
Panel A: All matched contracts				
Post	0.0010	0.0003	0.0197	0.0047
	(0.0004)	(0.0004)	(0.0014)	(0.0008)
Pre-event mean	0.4564	0.0071	0.3728	-0.0197
Observations	3,105	3,105	2,565	2,565
Matched contracts	105	105	57	57
Panel B: Resolution-risk-exposed contracts				
Post	0.0117	0.0012	0.0204	0.0147
	(0.0031)	(0.0027)	(0.0016)	(0.0013)
Pre-event mean	0.1969	0.0356	0.3437	-0.0596
Observations	355	355	817	817
Matched contracts	12	12	19	19
Contract fixed effects	Yes	Yes	Yes	Yes
Polymarket-volume weights	Yes	Yes	Yes	Yes

Notes: Each observation is an as-of price snapshot for a deterministic Polymarket–Kalshi contract match. The dependent variables are Polymarket's absolute distance from 0.5 and that distance minus the matched Kalshi contract's absolute distance from 0.5. The reported coefficient is from a weighted regression of the dependent variable on a post indicator and matched-contract fixed effects. Parentheses contain the conventional standard errors used in the empirical analysis. They are not clustered and should not be used for formal inference. No significance stars are reported. The resolution-risk-exposed sample contains elections, office exits, leader contacts, and policy actions.

or in markets whose rules contain genuine interpretive ambiguity. It may also appear in spreads, depth, withdrawal of liquidity, or trading activity rather than in price attenuation. Conversely, the positive coefficients may simply reflect stale prices, ordinary event news, compositional changes, or the mechanical behavior of absolute-value outcomes. The present estimates cannot distinguish these explanations.

7 Limitations and Required Empirical Work

The analysis has four central limitations.

Data frequency and market quality. Last-trade snapshots are not synchronized executable prices. In the constructed panels, Kalshi observations are substantially staler than Polymarket observations, with long tails of contracts that have not traded for days. The analysis should impose maximum trade-age restrictions and, where available, use synchronized bid and ask quotes. Volume and other market-quality outcomes require genuine event-time series rather than static snapshots.

Matching and contract equivalence. Deterministic title parsing does not compare full rule text, and exact normalized keys can hide economically important differences. Accepted matches should be checked against full resolution rules, especially the resolution source, deadline, threshold, and cancellation provisions. The segmented retrieval-plus-language-model procedure described above could increase coverage while retaining human validation of the final sample.

Event definition and windows. The one-day mineral-rights window and multi-day Zelensky window are not directly comparable. The analysis should document the key proposal, dispute, vote, resolution, and public-reporting timestamps, then report event-time estimates under narrower and wider windows.

Inference. The reported standard errors do not account adequately for repeated observations within a contract. Inference should cluster by matched contract, and the small number of controversy events means the evidence should remain descriptive. Results should also be reported unweighted and with predetermined pre-event weights because ex-post Polymarket volume may respond to the controversy itself.

8 Conclusion

The project has established a useful data and matching infrastructure for studying institutional risk in prediction markets. It has not yet established that UMA resolution controversies caused a broad loss of price informativeness. In the matched sample, Polymarket prices generally become more, rather than less, extreme after the two episodes, including relative to Kalshi after the Zelensky-suit controversy. These findings are worth reporting precisely because they discipline the theory: a general attenuation-toward-0.5 mechanism is too simple for the available data. Further work should focus on contract validation, synchronized market data, event timing, and defensible inference.

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